Fixed-Shift Local Models within the Square-Root Distribution Range: Quarter-Scale Row Masks, Ramanujan Spectral Calibration, and a Vaughan Type-II Reduction

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Abstract

Let $u(n) = \Lambda(n) - 1$, fix a nonzero integer shift h, and let $W \in C_c^\infty((0,\infty);\mathbb{R})$. We develop two fixed-shift local models for the factor-coordinate program of TPC-10–TPC-14.

First, on the primitive radial shell

$$\mathcal{K}_h(X) = \sum_{\substack{a>1,\ b\geq 1 \\ (a,b)=1}} u(a)(\Lambda(ab+h) - 1)W(ab/X),$$

the correct row density is $\rho_h(a) = \mathbf{1}_{(a,h)=1}a/\varphi(a)$. Every bounded source-factor row mask supported on $a \leq X^{1/4-\varepsilon}$ has arbitrary logarithmic cancellation after centering by $\rho_h(a)$. In contrast, the complete row-centered residual retains the explicit main term

$$\left(\frac{\varphi(|h|)}{|h|} - \beta_h\right) X I_0(W) \log X, \qquad \beta_h = \prod_{p|h}\left(1 - \frac{p}{p^2 - p + 1}\right).$$

For $h = 2$ the coefficient is $1/6$. When $I_0(W) \neq 0$, the residual logarithmic main term is therefore rigorously forced beyond the quarter-scale Type-I region; it is not an endpoint prime-pair estimate.

Second, define the finite Heath–Brown Ramanujan model and its residual by

$$\Lambda_Q(n) = \sum_{q\leq Q} \frac{\mu(q)}{\varphi(q)} c_q(n), \qquad r_Q(n) = \Lambda(n) - \Lambda_Q(n).$$

Its rational translation spectrum has Farey spacing at least Q^{-2}, and its fixed-shift autocorrelation is the truncated singular series. For fixed $0 < \delta < 1/2$ and $Q = X^{1/2-\delta}$, maximal Bombieri–Vinogradov and the finite spectral large sieve give

$$\sum_n \Lambda(n)\Lambda(n+h)W(n/X) = \mathfrak{S}(h)XI_0(W) + \sum_n r_Q(n)r_Q(n+h)W(n/X) + O_A(X(\log X)^{-A}).$$

Moreover, all bounded fixed-shift Type-I row masks up to modulus Q are controlled after this Ramanujan calibration. Applying Vaughan's identity with $U = V = X^{1/4-\delta/2}$ then discharges every Type-I packet in that decomposition and yields an exact reduction of the remaining error to one explicitly displayed hard bilinear packet. For an admissible even h with $I_0(W) \neq 0$, the Hardy–Littlewood fixed-shift asymptotic is equivalent to an $o(X)$ estimate for that packet. We do not estimate it, prove a twin-prime lower bound, enlarge the level of distribution, or claim a breach of sieve parity.

Keywords: fixed-shift prime correlations; Ramanujan sums; Heath–Brown model; Bombieri–Vinogradov theorem; Vaughan identity; Type-II reduction; sieve parity.

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1 Introduction and scope

The preceding factor-coordinate papers separated two losses that are often conflated. At a prescribed shift, TPC-14 evaluates the primitive radial shell only after summing all orientations. After a growing short-shift mean, it controls every bounded orientation mask, including the endpoint mask, but the shift coordinate is then averaged [14]. The remaining target was stated there in analytic rather than algebraic terms: subtract the correct local main term at fixed h, discharge the Type-I region, and identify the bilinear packet that still has to be estimated.

This paper carries out that reduction. All arithmetic inputs are classical: maximal Bombieri–Vinogradov, the finite Ramanujan model used by Heath–Brown, a separated-frequency mean-value inequality, and Vaughan's identity [1, 6, 11, 12]. The point is their label-preserving assembly at one fixed shift. No new distribution theorem for primes is assumed.

Fix throughout a nonzero integer h and a real $W \in C_c^\infty((0,\infty))$. Choose $0 < \alpha < \beta$ with

$$\operatorname{supp} W \subset [\alpha, \beta], \tag{1}$$

and put

$$I_0(W) = \int_0^\infty W(u)\,du, \qquad I_1(W) = \int_0^\infty W(u)\log u\,du. \tag{2}$$

All asymptotics are as $X \to \infty$ with h and W fixed. The von Mangoldt function is extended by zero to nonpositive integers.

1.1 The shell-local density and the quarter-scale theorem

Write

$$u(n) = \Lambda(n) - 1, \qquad \rho_h(a) = \mathbf{1}_{(a,h)=1}\frac{a}{\varphi(a)}. \tag{3}$$

The exact $k = 1$ shell from TPC-14 is

$$\mathcal{K}_h(X) = \sum_{\substack{a>1,\ b\geq 1\\ (a,b)=1}} u(a)(\Lambda(ab+h) - 1)W(ab/X). \tag{4}$$

For a row with source factor a, the target $ab + h$ lies in a prime progression. Its conditional density is $\rho_h(a)$, not 1. We split

$$\mathcal{K}_h(X) = \mathcal{M}_h(X) + \mathcal{R}_h(X), \tag{5}$$

$$\mathcal{M}_h(X) = \sum_{\substack{a>1,\ b\geq 1\\ (a,b)=1}} u(a)(\rho_h(a) - 1)W(ab/X), \tag{6}$$

$$\mathcal{R}_h(X) = \sum_{\substack{a>1,\ b\geq 1\\ (a,b)=1}} u(a)(\Lambda(ab+h) - \rho_h(a))W(ab/X). \tag{7}$$

Let

$$\beta_h = \prod_{p\mid h}\left(1 - \frac{p}{p^2 - p + 1}\right), \quad \kappa_h = \frac{1}{\zeta(2)} - \frac{\varphi(|h|)}{|h|}, \quad \delta_h = \frac{\varphi(|h|)}{|h|} - \beta_h. \tag{8}$$

We prove

$$\mathcal{M}_h(X) = \kappa_h X I_0(W)\log X + O_{h,W}(X), \qquad \mathcal{R}_h(X) = \delta_h X I_0(W)\log X + O_{h,W}(X). \tag{9}$$

For $|h|>1$, $\delta_h>0$; for $h=2$,

$$\kappa_2=\frac{6}{\pi^2}-\frac{1}{2},\qquad \delta_2=\frac{1}{6}. \tag{10}$$

The positive fixed-shift localization is uniform over a complete bounded row-mask ball. For every $A>0$ there is $B>0$ such that, whenever

$$Y^2\le X^{1/2}(\log X)^{-B}, \tag{11}$$

one has uniformly for $|\omega_a|\le 1$

$$\sum_{2\le a\le Y}\omega_a u(a)\sum_{\substack{b\ge 1\\(a,b)=1}}(\Lambda(ab+h)-\rho_h(a))W(ab/X)\ll_A X(\log X)^{-A}. \tag{12}$$

The square on Y is genuine: inserting $(a,b)=1$ creates moduli $ad\le a^2$. Combining (12) with (9) shows, when $\delta_h I_0(W)\ne 0$, that the $\delta_h X I_0(W)\log X$ residual lies beyond this quarter-scale Type-I region. It does not show that any one orientation, especially $b=1$, carries positive mass.

1.2 The finite Ramanujan local factor

Let

$$c_q(n)=\sum_{\substack{a \bmod q\\(a,q)=1}}\mathrm{e}^{2\pi i a n/q},\qquad \Lambda_Q(n)=\sum_{q\le Q}\frac{\mu(q)}{\varphi(q)}c_q(n),\qquad r_Q(n)=\Lambda(n)-\Lambda_Q(n). \tag{13}$$

This is the finite Heath–Brown Ramanujan model. The reduced fractions a/q, $q\le Q$, are the eigenfrequencies of translation on its finite periodic factor; distinct frequencies are separated by at least Q^{-2}. Its fixed-shift correlation is

$$\mathfrak{S}_Q(h)=\sum_{q\le Q}\frac{\mu(q)^2}{\varphi(q)^2}c_q(h). \tag{14}$$

For fixed h, this converges absolutely to the prime-pair singular series

$$\mathfrak{S}(h)=\sum_{q\ge 1}\frac{\mu(q)^2}{\varphi(q)^2}c_q(h). \tag{15}$$

Thus $\mathfrak{S}(h)=0$ for odd h, while for even h

$$\mathfrak{S}(h)=2\prod_{p>2}\left(1-\frac{1}{(p-1)^2}\right)\prod_{\substack{p\mid h\\p>2}}\frac{p-1}{p-2}. \tag{16}$$

For fixed $0<\delta<1/2$, put

$$Q=\lfloor X^{1/2-\delta}\rfloor. \tag{17}$$

The finite-spectrum mean value theorem and maximal Bombieri–Vinogradov give, for every $A>0$,

$$\sum_n \Lambda_Q(n)\Lambda_Q(n+h)W(n/X)=\mathfrak{S}_Q(h)XI_0(W)+O_A(X(\log X)^{-A}), \tag{18}$$

$$\sum_n \Lambda(n)\Lambda_Q(n+h)W(n/X)=\mathfrak{S}_Q(h)XI_0(W)+O_A(X(\log X)^{-A}), \tag{19}$$

and the analogous reversed cross term. Consequently

$$\sum_n \Lambda(n)\Lambda(n+h)W(n/X)=\mathfrak{S}(h)XI_0(W)+\sum_n r_Q(n)r_Q(n+h)W(n/X)+O_A(X(\log X)^{-A}). \tag{20}$$

This is a local-factor decomposition, not a bound for its residual term.

1.3 The fixed-shift Type-II reduction

The model is calibrated on every progression modulus $m \le Q$:

$$\lim_{J\to\infty} \frac{1}{J} \sum_{j\le J} \Lambda_Q(mj+h) = \rho_h(m). \tag{21}$$

The equality uses the complete divisor set $q \mid m$; it can fail if $m > Q$. Quantitatively, maximal Bombieri–Vinogradov and finite-period summation imply the full fixed-shift Type-I mask estimate

$$\sup_{|\omega_m|\le 1} \left| \sum_{m\le Q} \omega_m \sum_j r_Q(mj+h) W(mj/X) \right| \ll_A X(\log X)^{-A}. \tag{22}$$

Let

$$U = V = \lfloor Q^{1/2} \rfloor = X^{1/4-\delta/2} + O(1), \qquad \beta_V(k) = \sum_{\substack{d\mid k\\ d\le V}} \mu(d). \tag{23}$$

Then $UV \le Q$. Applying Vaughan's identity to the one-sided residual form and using (22) gives

$$\begin{aligned}&\sum_n \Lambda(n)\Lambda(n+h)W(n/X)\\ &\quad = \mathfrak{S}(h) X I_0(W) + \mathcal{B}_{h,\delta}(X) + O_A(X(\log X)^{-A}),\end{aligned} \tag{24}$$

$$\mathcal{B}_{h,\delta}(X) := -\sum_{\ell>U,\ k>V} \Lambda(\ell)\beta_V(k) r_Q(\ell k+h) W(\ell k/X). \tag{25}$$

On the support, both factors lie in the corridor

$$X^{1/4-\delta/2} < \ell, k \ll_W X^{3/4+\delta/2}. \tag{26}$$

For an admissible even h with $I_0(W) \neq 0$, the weighted Hardy–Littlewood asymptotic is therefore equivalent to $\mathcal{B}_{h,\delta}(X) = o(X)$. This identifies the remaining Type-II gate exactly. The paper proves no estimate of that strength for $\mathcal{B}_{h,\delta}$.

1.4 Claim ledger

Object	Result here	Strict boundary
Primitive $k=1$ shell	Fixed-h row centering and all bounded masks through the quarter scale	Orientations b are still aggregated outside the mask variable
Finite Ramanujan factor	Exact translation spectrum and singular-series main term through $Q = X^{1/2-\delta}$	The residual correlation is not bounded
Type-I progression rows	Full bounded-mask closure at one prescribed h	Calibration requires every row modulus to be at most Q
Vaughan decomposition	All Type-I packets produced here are discharged; one explicit hard packet remains	No fixed-h Type-II cancellation is proved
Endpoint $h=2$	Exact equivalence with the hard residual packet	No twin-prime asymptotic or lower bound

The distinction in the last row is essential. For nonnegative W, even a natural-scale positive lower bound for the $h=2$ prime-pair correlation would already imply infinitely many twin primes. Nothing of that kind is deduced below.

2 The row-local model on the primitive shell

Throughout this section, $h \neq 0$ is fixed. Put

$$\rho_h(a) := \mathbf{1}_{(a,h)=1} \frac{a}{\varphi(a)}, \qquad a \geq 1. \tag{27}$$

This is the local von Mangoldt density on the row $ab+h$ after b is restricted by $(a,b)=1$. If $(a,h)>1$, every value $ab+h$ has a fixed prime divisor, apart from the exceptional possibility that it is a power of that prime; the appropriate main density is therefore zero. If $(a,h)=1$, inclusion–exclusion over $(a,b)=1$ gives density $a/\varphi(a)$ rather than the ambient density one.

Recall $u(a) = \Lambda(a) - 1$. Decompose the centered primitive shell as

$$\mathcal{K}_h(X) := \sum_{\substack{a>1,\ b\geq 1\\ (a,b)=1}} u(a)\big(\Lambda(ab+h)-1\big)W(ab/X), \tag{28}$$

$$\mathcal{M}_h(X) := \sum_{\substack{a>1,\ b\geq 1\\ (a,b)=1}} u(a)\big(\rho_h(a)-1\big)W(ab/X), \tag{29}$$

$$\mathcal{R}_h(X) := \sum_{\substack{a>1,\ b\geq 1\\ (a,b)=1}} u(a)\big(\Lambda(ab+h)-\rho_h(a)\big)W(ab/X). \tag{30}$$

Thus, exactly at every finite X,

$$\mathcal{K}_h(X) = \mathcal{M}_h(X) + \mathcal{R}_h(X). \tag{31}$$

Define

$$\kappa_h := \frac{1}{\zeta(2)} - \frac{\varphi(|h|)}{|h|}, \qquad \delta_h := \frac{\varphi(|h|)}{|h|} - \beta_h. \tag{32}$$

The two constants add to the coefficient in the fixed-shell theorem of TPC-14:

$$\kappa_h + \delta_h = \frac{1}{\zeta(2)} - \beta_h. \tag{33}$$

Moreover,

$$\beta_h = \frac{\varphi(|h|)}{|h|} \prod_{p|h} \left(1 - \frac{1}{p^2-p+1}\right). \tag{34}$$

Hence $\delta_h > 0$ whenever $|h|>1$, while $\delta_1 = \delta_{-1} = 0$.

The main term of $\mathcal{M}_h$ is not obtained by applying Euler summation to each row and then summing the individual endpoint errors. That procedure is too wasteful when a is as large as X. We instead group by the product and compute the relevant Dirichlet series.

For $r \geq 1$, put

$$m_h(r) := \sum_{\substack{a\| r\\ a>1}} u(a)\big(\rho_h(a)-1\big), \tag{35}$$

where $a \,\|\, r$ means that $a \mid r$ and $(a, r/a) = 1$. Then

$$\mathcal{M}_h(X) = \sum_{r\geq 1} m_h(r) W(r/X). \tag{36}$$

Lemma 2.1 (Dirichlet factorization of the row model)**.** *For* $\Re s > 1$,

$$\sum_{r\geq 1} \frac{m_h(r)}{r^s} = \zeta(s)^2 \left(\frac{1}{\zeta(2s)} - C_h(s)\right) + \zeta(s) L_h(s), \tag{37}$$

where

$$C_h(s) := \prod_{p\mid h}(1-p^{-s})\prod_{p\nmid h}(1-p^{-s})\left(1+\frac{p^{1-s}}{p-1}\right), \tag{38}$$

$$L_h(s) := \sum_{p\nmid h}\frac{\log p}{p-1}p^{-s} - \sum_{p\mid h}(\log p)p^{-s}. \tag{39}$$

Both C_h and L_h are holomorphic in a neighborhood of $s=1$, and

$$C_h(1) = \frac{\varphi(|h|)}{|h|}. \tag{40}$$

Consequently, the double-pole coefficient at $s=1$ in (37) is κ_h.

Proof. For fixed a one has

$$\sum_{\substack{b\geq 1\\(a,b)=1}} b^{-s} = \zeta(s)\prod_{p\mid a}(1-p^{-s}).$$

Since $\rho_h(1)=1$, the formally missing term $a=1$ contributes zero. It follows from (35) that the left side of (37) equals

$$\zeta(s)\sum_{a\geq 1}\frac{(\Lambda(a)-1)(\rho_h(a)-1)}{a^s}\prod_{p\mid a}(1-p^{-s}). \tag{41}$$

We first treat the contribution of the constant part -1 in $\Lambda-1$. The two multiplicative series appearing there are

$$\sum_{a\geq 1}\frac{1}{a^s}\prod_{p\mid a}(1-p^{-s}) = \prod_p(1+p^{-s}) = \frac{\zeta(s)}{\zeta(2s)},$$

$$\sum_{a\geq 1}\frac{\rho_h(a)}{a^s}\prod_{p\mid a}(1-p^{-s}) = \prod_{p\nmid h}\left(1+\frac{p^{1-s}}{p-1}\right) = \zeta(s)C_h(s).$$

For $p\nmid h$, the corresponding local factor in C_h is

$$(1-p^{-s})\left(1+\frac{p^{1-s}}{p-1}\right) = 1+\frac{p^{-s}}{p-1}-\frac{p^{1-2s}}{p-1}.$$

The product therefore converges normally on compact subsets of a half-plane containing $s=1$. At $s=1$ this local factor is exactly one. Only the finitely many primes dividing h remain, which proves (40).

For the Λ part, only prime powers occur. If $p\nmid h$, their total contribution to the inner series in (41) is

$$\sum_{j\geq 1}(\log p)\left(\frac{p}{p-1}-1\right)(1-p^{-s})p^{-js} = \frac{\log p}{p-1}p^{-s}.$$

If $p\mid h$, it is instead $-(\log p)p^{-s}$. This gives (39). The first prime sum is normally convergent for $\Re s>0$, locally around $s=1$, and the second is finite. Multiplying the resulting inner identity by the outer $\zeta(s)$ in (41) proves (37). Its only double pole at $s=1$ comes from the first term, with coefficient $1/\zeta(2)-C_h(1)=\kappa_h$. □

Theorem 2.2 (Global row-model law)**.** *For every fixed nonzero h,*

$$\mathcal{M}_h(X) = \kappa_h X I_0(W)\log X + O_{h,W}(X). \tag{42}$$

Proof. The Euler products and prime series in lemma 2.1 converge normally for $\Re s > 1/2$ and $\Re s > 0$, respectively. Since $1/\zeta(2s)$ is holomorphic for $\Re s > 1/2$, the factorization continues meromorphically to that half-plane, with its only pole on $\Re s \geq 1$ at $s = 1$. It is a double pole with leading coefficient κ_h; all remaining terms have at most a simple pole there.

Let $\widehat{W}(s) = \int_0^\infty W(x)x^{s-1}\,dx$. Mellin inversion on $\Re s > 1$, followed by a shift to $\Re s = 1/2+\varepsilon$, is legitimate: $\widehat{W}$ decays faster than every power in vertical strips, while the other factors have polynomial vertical growth. The residue at $s = 1$ therefore gives

$$\sum_r m_h(r)W(r/X) = \kappa_h X I_0(W)\log X + c_{h,W}X + o_{h,W}(X)$$

for a finite secondary constant $c_{h,W}$, with an error $O_{h,W,\varepsilon}(X^{1/2+\varepsilon})$. Absorbing the secondary term and using (36) proves the theorem. □

TPC-14 [14] evaluates the original shell, for the same fixed h, as

$$\mathcal{K}_h(X) = \left(\frac{1}{\zeta(2)} - \beta_h\right) X I_0(W)\log X + O_{h,W}(X). \tag{43}$$

Subtracting the row model isolates what is not explained by local row density.

Corollary 2.3 (Global row-residual law)**.** *For every fixed nonzero h,*

$$\mathcal{R}_h(X) = \delta_h X I_0(W)\log X + O_{h,W}(X). \tag{44}$$

In particular,

$$\kappa_2 = \frac{6}{\pi^2} - \frac{1}{2}, \qquad \delta_2 = \frac{1}{6}. \tag{45}$$

Proof. Use the exact identity (31), then subtract (42) from (43). The coefficient is

$$\left(\frac{1}{\zeta(2)} - \beta_h\right) - \kappa_h = \frac{\varphi(|h|)}{|h|} - \beta_h = \delta_h.$$

For $h = 2$, one has $\varphi(2)/2 = 1/2$ and $\beta_2 = 1/3$. □

Remark 2.4 (What the residual law says). The quantity $\mathcal{R}_h$ is still summed over every primitive orientation b. Equation (44) says that local row-density calibration does not explain the full fixed-shell main term. It neither assigns the remaining term to any individual orientation nor gives information about the endpoint $b = 1$.

3 Quarter-scale row masks at a fixed shift

The global residual in (44) cannot be distributed among individual orientations by the fixed-shell theorem. There is, however, a full bounded-mask theorem on the one-sided Type-I range. The loss from the square-root modulus range to the quarter-scale source-factor row range is caused by the coprimality condition $(a, b) = 1$.

We first record the precise weighted form of Bombieri–Vinogradov that is used below. For $q \geq 1$, define the smooth row discrepancy

$$E_h(q; X) := \sum_{j\geq 1} \Lambda(qj + h)W(qj/X) - \mathbf{1}_{(q,h)=1}\frac{X I_0(W)}{\varphi(q)}. \tag{46}$$

Lemma 3.1 (Divisor-weighted maximal Bombieri–Vinogradov)**.** *Fix an integer $J \geq 0$. For every $A > 0$ there is $B = B(A, J, h, W) > 0$ such that, whenever*

$$Q \leq \frac{X^{1/2}}{(\log X)^B}, \tag{47}$$

one has

$$\sum_{q \leq Q} (\log(2q))^J \tau(q)^J |E_h(q; X)| \ll_{A,J,h,W} \frac{X}{(\log X)^A}. \tag{48}$$

The same conclusion holds with the displayed weight replaced by any smaller nonnegative weight.

Proof. For $(q, h) = 1$, smooth partial summation bounds $|E_h(q; X)|$ by a fixed multiple of

$$\max_{y \ll_{h,W} X} \left| \sum_{\substack{n \leq y \\ n \equiv h \pmod q}} \Lambda(n) - \frac{y}{\varphi(q)} \right|.$$

The maximal Bombieri–Vinogradov theorem controls the sum of these quantities over $q \leq Q$ with an arbitrarily large fixed logarithmic saving [7, 9].

We explain why the divisor weight in (48) is legitimate. It is not obtained by multiplying the unweighted theorem by the maximum of $\tau(q)^J$. Choose a large fixed K. On the set $\tau(q)^J \leq (\log X)^K$, apply maximal Bombieri–Vinogradov with $A + K + J + 3$ in place of A. On the complementary set use the elementary bound

$$|E_h(q; X)| \ll_{h,W} \frac{X \log(2X)}{q} \qquad (q \leq X^{1/2}), \tag{49}$$

where the main term is absorbed using $q/\varphi(q) \ll \log(2q)$. If $J = 0$, the complementary set is empty. If $J \geq 1$, a fixed divisor moment handles it: more explicitly, on that set

$$\tau(q)^J \leq (\log X)^{-K} \tau(q)^{2J},$$

while, for a constant C_J depending only on J,

$$\sum_{q \leq Q} \frac{\tau(q)^{2J}}{q} \ll_J (\log(2Q))^{C_J}.$$

Taking $K > C_J + A + 2J + 4$ makes this contribution admissible. This proves the divisor-weighted extension in the reduced residue classes.

It remains to justify the zero main term when $(q, h) > 1$. If $\Lambda(qj + h) \neq 0$, then $qj + h = p^\nu$ for a prime $p \mid h$. On the support of W, only $O_{h,W}(1)$ exponents ν occur for each of the finitely many $p \mid h$. Moreover, $q \mid p^\nu - h$. The standard divisor bound therefore makes the total contribution of all such q, even with the weight in (48), $O_{J,h,W,\epsilon}(X^\epsilon)$ for every $\epsilon > 0$. This is smaller than the asserted bound. The lemma follows. $\square$

Let $Y \geq 2$, and let $\omega = (\omega_a)_{2 \leq a \leq Y}$ be an arbitrary complex mask satisfying

$$|\omega_a| \leq 1. \tag{50}$$

Define its row-centered primitive-shell statistic by

$$\mathcal{R}^{\omega}_{h,\leq Y}(X) := \sum_{\substack{2 \leq a \leq Y,\ b \geq 1 \\ (a,b)=1}} \omega_a u(a)(\Lambda(ab + h) - \rho_h(a)) W(ab/X). \tag{51}$$

Theorem 3.2 (Uniform quarter-scale row-mask transfer)**.** *For every $A > 0$ there is $B = B(A, h, W) > 0$ such that, uniformly over every mask satisfying (50) and every Y with*

$$Y^2 \leq \frac{X^{1/2}}{(\log X)^B}, \tag{52}$$

one has

$$\mathcal{R}^{\omega}_{h,\leq Y}(X) \ll_{A,h,W} \frac{X}{(\log X)^A}. \tag{53}$$

The implied constant is independent of Y and of the mask.

Proof. Insert

$$\mathbf{1}_{(a,b)=1} = \sum_{\substack{d|a\\ d|b}} \mu(d) \tag{54}$$

into the prime part of (51). After writing $b = dj$, it becomes

$$\sum_{a\leq Y} \omega_a u(a) \sum_{d|a} \mu(d) \sum_{j\geq 1} \Lambda(adj + h)W(adj/X), \tag{55}$$

where, if convenient, we set $\omega_1 = 0$ and retain the value $a = 1$. Every resulting modulus is

$$q = ad \leq Y^2.$$

Grouping (55) by q produces the coefficient

$$\gamma_q = \sum_{\substack{a\leq Y,\ d|a\\ ad=q}} \omega_a u(a)\mu(d), \qquad |\gamma_q| \leq (1 + \log(2q))\tau(q). \tag{56}$$

Thus lemma 3.1, with $J = 2$ after increasing B, replaces every inner prime sum by its reduced-class main term with total error $O_A(X(\log X)^{-A})$. If $(a, h) = 1$, every ad with $d \mid a$ is also coprime to h, and the elementary multiplicative identity

$$\sum_{d|a} \frac{\mu(d)}{\varphi(ad)} = \frac{1}{a} \tag{57}$$

applies. Indeed, at $a = p^{\nu}$ its left side is

$$\frac{1}{\varphi(p^{\nu})} - \frac{1}{\varphi(p^{\nu+1})} = \frac{1}{p^{\nu}}.$$

If $(a, h) > 1$, every modulus ad is nonreduced and its main term is zero; the prime-power exceptions have already been included in the proof of lemma 3.1. Consequently (55) equals

$$XI_0(W) \sum_{\substack{a\leq Y\\ (a,h)=1}} \frac{\omega_a u(a)}{a} + O_{A,h,W}(X(\log X)^{-A}). \tag{58}$$

For the local model, another use of (54) and Euler summation gives, uniformly in a,

$$\sum_{\substack{b\geq 1\\ (a,b)=1}} W(ab/X) = \frac{X\varphi(a)}{a^2} I_0(W) + O_W(\tau(a)). \tag{59}$$

Multiplication by $\rho_h(a)$ turns its main term into $\mathbf{1}_{(a,h)=1} XI_0(W)/a$, exactly matching (58). The accumulated Euler error is

$$\ll_W \sum_{a\leq Y} |u(a)| \frac{a}{\varphi(a)} \tau(a) \ll_W Y(\log(2Y))^C$$

for an absolute fixed C. Under (52), this is $O_A(X(\log X)^{-A})$ for every fixed A. Subtraction proves (53). □

The theorem immediately evaluates the original, ambiently centered truncation. Put

$$\mathcal{K}^{\omega}_{h,\le Y}(X) := \sum_{\substack{2\le a\le Y,\ b\ge 1\\ (a,b)=1}} \omega_a u(a)(\Lambda(ab+h)-1)W(ab/X), \tag{60}$$

$$\mathcal{M}^{\omega}_{h,\le Y}(X) := \sum_{\substack{2\le a\le Y,\ b\ge 1\\ (a,b)=1}} \omega_a u(a)(\rho_h(a)-1)W(ab/X). \tag{61}$$

Corollary 3.3 (Original truncated shell with a bounded row mask)**.** *Under the hypotheses of theorem 3.2,*

$$\begin{aligned}\mathcal{K}^{\omega}_{h,\le Y}(X) = XI_0(W)\sum_{2\le a\le Y}\omega_a u(a)\left(\frac{\mathbf{1}_{(a,h)=1}}{a}-\frac{\varphi(a)}{a^2}\right)&\\ + O_{A,h,W}(X(\log X)^{-A}).&\end{aligned} \tag{62}$$

Proof. The exact identity

$$\mathcal{K}^{\omega}_{h,\le Y} = \mathcal{M}^{\omega}_{h,\le Y} + \mathcal{R}^{\omega}_{h,\le Y}$$

and theorem 3.2 reduce the problem to the model. Apply (59); since

$$(\rho_h(a)-1)\frac{\varphi(a)}{a^2} = \frac{\mathbf{1}_{(a,h)=1}}{a} - \frac{\varphi(a)}{a^2},$$

its main term is the displayed finite sum. The accumulated Euler error is bounded exactly as in the proof of the theorem and is absorbed after increasing B. □

For the unmasked truncation, the finite arithmetic sum in (62) has a simple asymptotic.

Lemma 3.4 (Mean of the row-model coefficient)**.** *Uniformly for $y\ge 2$,*

$$\sum_{2\le a\le y} u(a)\left(\frac{\mathbf{1}_{(a,h)=1}}{a}-\frac{\varphi(a)}{a^2}\right) = \kappa_h \log y + O_h(1). \tag{63}$$

Proof. The constant part of $u=\Lambda-1$ contributes

$$\sum_{a\le y}\frac{\varphi(a)}{a^2} - \sum_{\substack{a\le y\\ (a,h)=1}}\frac{1}{a} = \left(\frac{1}{\zeta(2)} - \frac{\varphi(|h|)}{|h|}\right)\log y + O_h(1).$$

For the Λ part, write $a=p^j$. If $p\nmid h$, the summand is $(\log p)/p^{j+1}$; if $p\mid h$, it is $-(p-1)(\log p)/p^{j+1}$. The resulting prime-power series converges absolutely, so it contributes $O_h(1)$. This proves the lemma. □

Corollary 3.5 (Quarter-scale truncation and residual escape)**.** *Suppose (52) holds and take $\omega_a=1$. Then*

$$\mathcal{K}_{h,\le Y}(X) = \kappa_h X I_0(W)\log Y + O_{h,W}(X), \tag{64}$$

$$\mathcal{R}_{h,\le Y}(X) \ll_{A,h,W} X(\log X)^{-A}. \tag{65}$$

If $\mathcal{K}_{h,>Y}$, $\mathcal{M}_{h,>Y}$, and $\mathcal{R}_{h,>Y}$ denote the corresponding sums with $a>Y$, then

$$\mathcal{R}_{h,>Y}(X) = \delta_h X I_0(W)\log X + O_{h,W}(X), \tag{66}$$

$$\mathcal{K}_{h,>Y}(X) = \left(\frac{1}{\zeta(2)}-\beta_h\right)XI_0(W)\log X - \kappa_h XI_0(W)\log Y + O_{h,W}(X). \tag{67}$$

In particular, for $h=2$ the row-centered residual remaining beyond the quarter-scale range has leading coefficient $\delta_2 = 1/6$.

Proof. Equation (64) follows from corollary 3.3 and lemma 3.4; (65) is theorem 3.2. Subtract the latter from the global law (44) to obtain (66). Finally subtract (64) from (43) to obtain (67). □

Remark 3.6 (Strict interpretation of escape). The word "escape" refers only to the a-coordinate: every bounded mask on $a \le Y$ sees logarithmic cancellation after the correct row centering, while the complete residual has the main term in (44). The tail $a > Y$ still contains all allowed b, and (66) does not identify the endpoint $b = 1$, a balanced Type-II sector, or any individual orientation. It gives no fixed-shift prime-pair asymptotic or twin-prime lower bound.

4 The finite Ramanujan model as a rational translation spectrum

Write $\mathrm{e}(x) = \exp(2\pi i x)$. For $q \ge 1$, the Ramanujan sum is

$$c_q(n) = \sum_{\substack{a \pmod q \\ (a,q)=1}} \mathrm{e}(an/q) = \sum_{d|(q,n)} d\mu(q/d). \tag{68}$$

For $Q \ge 1$, define the finite Heath–Brown model and its residual by

$$\Lambda_Q(n) = \sum_{q \le Q} \frac{\mu(q)}{\varphi(q)} c_q(n), \qquad r_Q(n) = \Lambda(n) - \Lambda_Q(n). \tag{69}$$

The adjective "model" is important: Λ_Q is neither a pointwise minorant nor a pointwise approximation to Λ.

4.1 Translation spectrum and its resolution

Let

$$\mathscr{F}_Q = \left\{ \frac{a}{q} \pmod 1 : 1 \le q \le Q, (a,q) = 1 \right\}. \tag{70}$$

Every element is represented in lowest terms, so distinct points $\xi = a/q$ and $\xi' = b/r$ satisfy

$$\|\xi - \xi'\|_{\mathbb{R}/\mathbb{Z}} \ge \frac{1}{qr} \ge Q^{-2}. \tag{71}$$

Let $L_Q = \mathrm{lcm}(1, \ldots, Q)$. The functions below live on the finite cyclic group $\mathbb{Z}/L_Q\mathbb{Z}$. On the subspace spanned by $e_\xi(n) = \mathrm{e}(n\xi)$, let translation act by

$$T^h e_\xi = \mathrm{e}(h\xi) e_\xi. \tag{72}$$

Thus Λ_Q is a finite vector in a rational pure-point translation spectrum:

$$\Lambda_Q(n) = \sum_{\xi = a/q \in \mathscr{F}_Q} z_\xi \mathrm{e}(n\xi), \qquad z_{a/q} = \frac{\mu(q)}{\varphi(q)}. \tag{73}$$

This is a spectral calibration of local congruence densities, not a dynamical representation of the primes themselves.

We shall use the classical large-sieve inequality in the form

$$\sum_{n \in I} \left| \sum_{\xi \in \mathscr{F}_Q} z_\xi \mathrm{e}(n\xi) \right|^2 \le (|I| + Q^2) \sum_{\xi \in \mathscr{F}_Q} |z_\xi|^2, \tag{74}$$

up to an absolute change in the constant [8]. The Q^2 term is precisely the reciprocal gap in (71). In the range $Q = X^{1/2-\delta}$ it is $o(X)$ by a power. We also use the following smoothed consequence of the same spacing. If $W \in C_c^\infty((0,\infty))$, then for every $A > 0$ and fixed $\delta > 0$,

$$\begin{aligned}\sum_n \Lambda_Q(n)\Lambda_Q(n+h)W(n/X)\\ = XI_0(W)\sum_{q\le Q}\frac{\mu(q)^2 c_q(h)}{\varphi(q)^2} + O_{A,\delta,W}\big(X(\log X)^{-A}\big),\end{aligned} \tag{75}$$

uniformly for $Q \le X^{1/2-\delta}$. Indeed, insert (73). The zero frequency occurs precisely when the two reduced fractions have the same denominator and opposite numerators; its contribution is the displayed Ramanujan sum. Every other pair has distance at least Q^{-2} from an integer. Poisson summation gives

$$\sum_n W(n/X)\mathrm{e}(n\theta) \ll_{B,W} X(1+X\|\theta\|)^{-B}.$$

Since the ℓ^1 norm of the spectral coefficients is $O(Q)$, choosing B in terms of A and δ disposes of all nonzero frequencies. This argument also makes clear why a power gap between Q^2 and X is kept below.

4.2 The divisor expansion and exact progression means

The spectral expansion has a complementary divisor form that is better suited to incomplete arithmetic progressions.

Lemma 4.1 (Finite divisor expansion). *For every positive integer n,*

$$\Lambda_Q(n) = \sum_{\substack{d\mid n\\ d\le Q}} \lambda'_Q(d), \tag{76}$$

where

$$\begin{aligned}\lambda'_Q(d) &= d\sum_{r\le Q/d}\frac{\mu(dr)\mu(r)}{\varphi(dr)}\\ &= \mu(d)\frac{d}{\varphi(d)}\sum_{\substack{r\le Q/d\\ (r,d)=1}}\frac{\mu(r)^2}{\varphi(r)}\end{aligned} \tag{77}$$

when d is squarefree, and $\lambda'_Q(d) = 0$ otherwise. Uniformly in $Q \ge 2$,

$$|\lambda'_Q(d)| \ll \frac{d}{\varphi(d)}\log(2Q/d), \tag{78}$$

$$\sum_{d\le Q}|\lambda'_Q(d)| \ll Q(\log(2Q))^2. \tag{79}$$

Proof. Insert the divisor identity in (68) into (69), write $q = dr$, and interchange the finite sums. If $\mu(dr) \ne 0$, then d and r are squarefree and coprime, which gives the second line of (77). The standard bound

$$\sum_{r\le y}\frac{\mu(r)^2}{\varphi(r)} \ll \log(2y)$$

proves (78). Summing it, or interchanging the d, r sums once more and using $n/\varphi(n) \ll \log\log(3n)$, proves the deliberately nonoptimal but sufficient estimate (79). ☐

For fixed $h \neq 0$, put

$$\rho_h(m) = \mathbf{1}_{(m,h)=1} \frac{m}{\varphi(m)}. \tag{80}$$

Lemma 4.2 (Exact complete-period row mean)**.** *If $m \le Q$, then the mean of $j \mapsto \Lambda_Q(mj+h)$ over a complete period is exactly $\rho_h(m)$.*

Proof. From the exponential definition,

$$\mathbb{E}_j c_q(mj+h) = \begin{cases} c_q(h), & q \mid m, \\ 0, & q \nmid m. \end{cases} \tag{81}$$

Indeed, a reduced numerator a can make am/q integral only when $q \mid m$. Because $m \le Q$, every divisor $q \mid m$ occurs in the finite model. Hence the mean in question is

$$\sum_{q \mid m} \frac{\mu(q) c_q(h)}{\varphi(q)}.$$

This is multiplicative in m. If $p \mid m$ and $p \nmid h$, its local factor is $1 + 1/(p-1) = p/(p-1)$. If $p \mid (m,h)$, the local factor is $1 - 1 = 0$. Their product is (80). ☐

The condition $m \le Q$ is structural, not cosmetic. If a prime divisor of m exceeds Q, its local factor is absent from the finite model, so the asserted row mean need not equal $m/\varphi(m)$. This is the reason for the parameter inequality $UV \le Q$ in the Vaughan reduction below.

Lemma 4.3 (Smoothed incomplete-row error)**.** *Let $F \in C_c^1((0,\infty))$. Uniformly for $m \le Q$,*

$$\sum_{j \ge 1} \Lambda_Q(mj+h) F(mj/X) - \rho_h(m) \frac{X}{m} I_0(F) \ll_F Q(\log(2Q))^2, \tag{82}$$

$$\sum_{j \ge 1} (\log j) \Lambda_Q(mj+h) F(mj/X) - \rho_h(m) \frac{1}{m} \int_0^\infty (\log(t/m)) F(t/X)\, dt \ll_F Q(\log(2XQ))^3. \tag{83}$$

Proof. Use (76). For a fixed d, the congruence $d \mid mj+h$ is either insoluble or is one residue class modulo $d/(d,m)$. Euler summation on that lattice has an $O_F(1)$ error; with the multiplier $\log j$, partial summation costs $O_F(\log(2XQ))$. The continuous main terms recombine to the complete-period mean in lemma 4.2. Summing the boundary errors with (79) proves both estimates. ☐

The last lemma avoids a common false estimate. Taking absolute values separately over all reduced numerators in the Fourier expansion can lose Q^2 in one row. The divisor expansion retains the congruence cancellation and costs only the ℓ^1 norm in (79).

5 Cross-correlations and extraction of the singular series

Define the finite singular series

$$\mathfrak{S}_Q(h) = \sum_{q \le Q} \frac{\mu(q)^2 c_q(h)}{\varphi(q)^2}. \tag{84}$$

For fixed $h \neq 0$, the infinite series is absolutely convergent and equals the prime-pair singular series

$$\mathfrak{S}(h) = \sum_{q \ge 1} \frac{\mu(q)^2 c_q(h)}{\varphi(q)^2}. \tag{85}$$

Thus $\mathfrak{S}(h) = 0$ for odd h, while for even h

$$\mathfrak{S}(h) = 2 \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \prod_{\substack{p|h \\ p>2}} \frac{p-1}{p-2}. \tag{86}$$

Lemma 5.1 (Finite calibration and its tail)**.** *For fixed $h \neq 0$,*

$$\sum_{\substack{d \le Q \\ (d,h)=1}} \frac{\lambda'_Q(d)}{\varphi(d)} = \mathfrak{S}_Q(h), \tag{87}$$

and

$$\mathfrak{S}_Q(h) = \mathfrak{S}(h) + O_h\left(\frac{(\log(2Q))^2}{Q}\right). \tag{88}$$

Proof. For the first identity, insert (77) and reverse the finite divisor rearrangement. Equivalently, average $c_q(n+h)$ over the reduced residue classes occupied by a prime n. For squarefree q,

$$\sum_{\substack{a \pmod q \\ (a,q)=1}} c_q(a+h) = \mu(q) c_q(h), \tag{89}$$

which gives exactly (87) after multiplying by $\mu(q)/\varphi(q)^2$.

For the tail, write $g = (q,h)$. Since g ranges over the divisors of the fixed integer h, the standard estimates for $1/\varphi(n)$ reduce the absolute tail to $O_h(\sum_{r>Q/|h|} \varphi(r)^{-2})$, which is $O_h(Q^{-1}(\log(2Q))^2)$. The Euler product of (85) gives (86). $\square$

We now compare the model with the primes. We use the maximal Bombieri–Vinogradov theorem in its usual smooth form: for every $A > 0$ there is $B > 0$ such that, if

$$Q \le \frac{X^{1/2}}{(\log X)^B}, \tag{90}$$

then

$$\sum_{q \le Q} \max_{(a,q)=1} \max_{y \ll_W X} \left| \sum_{\substack{n \le y \\ n \equiv a \pmod q}} \Lambda(n) - \frac{y}{\varphi(q)} \right| \ll_{A,W} \frac{X}{(\log X)^A}. \tag{91}$$

This is a classical input [7, 9].

Theorem 5.2 (Both prime–model cross-correlations)**.** *Fix $h \neq 0$ and $W \in C_c^\infty((0,\infty))$. For every $A > 0$, after choosing B in (90) sufficiently large,*

$$\sum_n \Lambda(n)\Lambda_Q(n+h)W(n/X) = \mathfrak{S}_Q(h) X I_0(W) + O_{A,h,W}(X(\log X)^{-A}), \tag{92}$$

$$\sum_n \Lambda_Q(n)\Lambda(n+h)W(n/X) = \mathfrak{S}_Q(h) X I_0(W) + O_{A,h,W}(X(\log X)^{-A}). \tag{93}$$

Proof. For (92), use (76) to obtain progressions $n \equiv -h \pmod d$. If $(d,h) = 1$, smooth partial summation in (91) replaces the prime sum by $XI_0(W)/\varphi(d)$. The coefficient bound

$$|\lambda'_Q(d)| \ll (\log(2Q))^2 \tag{94}$$

follows from (78); requesting a larger exponent in Bombieri–Vinogradov absorbs this logarithm. The main terms give (87).

If $(d,h)>1$, the residue class is not reduced. Any von Mangoldt value in that class is a power of one of the finitely many primes dividing h. On the support of W, the total contribution is $O_{h,W,\varepsilon}(X^{\varepsilon})$: for each relevant prime power, sum (94) only over the divisors of a fixed integer. This is absorbed in the asserted error.

For (93), the congruence is $n+h\equiv h \pmod d$ with $d\mid n$. The same maximal theorem and the same non-reduced-class argument apply. The two main terms agree because the singular series is invariant under replacing h by $-h$. $\square$

Combining the cross-correlations with the model autocorrelation produces an exact residual identity at natural scale.

Theorem 5.3 (Ramanujan-calibrated correlation identity). *Fix $0<\delta<1/2$ and put $Q=\lfloor X^{1/2-\delta}\rfloor$. For every $A>0$,*

$$\sum_n \Lambda(n)\Lambda(n+h)W(n/X)$$
$$= \mathfrak{S}(h)XI_0(W)+\sum_n r_Q(n)r_Q(n+h)W(n/X)+O_{A,h,W,\delta}(X(\log X)^{-A}). \tag{95}$$

Proof. Expand $r_Q=\Lambda-\Lambda_Q$. By theorem 5.2, both cross terms equal $\mathfrak{S}_Q(h)XI_0(W)$ up to arbitrary logarithmic savings. By (75), the model autocorrelation has the same main term. The elementary identity

$$\Lambda\Lambda_h=\Lambda_Q(\Lambda_Q)_h+r_Q(\Lambda_Q)_h+\Lambda_Q(r_Q)_h+r_Q(r_Q)_h$$

therefore leaves one copy of $\mathfrak{S}_Q(h)XI_0(W)$. Finally use (88); its error is a power smaller than X. $\square$

Equation (95) is not an estimate for the residual autocorrelation. For an even admissible shift, proving that the last sum is $o(X)$ would already yield the Hardy–Littlewood asymptotic. The role of the finite model is to remove the correct local main term before the unresolved correlation is named.

6 Fixed-shift closure of the full Type-I row ball

The model is useful because its complete-period mean agrees exactly with the prime mean in every row whose modulus is visible at resolution Q. For $F\in C_c^1((0,\infty))$, define

$$R_{m,h;Q}(X;F)=\sum_{j\ge 1} r_Q(mj+h)F(mj/X), \tag{96}$$
$$R^{\log}_{m,h;Q}(X;F)=\sum_{j\ge 1}(\log j)r_Q(mj+h)F(mj/X). \tag{97}$$

Theorem 6.1 (Full bounded Type-I row masks). *Fix $h\neq 0$, $F\in C_c^1((0,\infty))$, and $\eta>0$. Let $M\le Q$ and suppose*

$$Q\le \frac{X^{1/2}}{(\log X)^B},\qquad MQ\le X^{1-\eta}. \tag{98}$$

For every $A>0$, if $B=B(A,h,F,\eta)$ is sufficiently large, then uniformly over all coefficient vectors $|\alpha_m|\le 1$,

$$\sup_{|\alpha_m|\le 1}\left|\sum_{m\le M}\alpha_m R_{m,h;Q}(X;F)\right| \ll_{A,h,F,\eta} X(\log X)^{-A}, \tag{99}$$

$$\sup_{|\alpha_m|\le 1}\left|\sum_{m\le M}\alpha_m R^{\log}_{m,h;Q}(X;F)\right| \ll_{A,h,F,\eta} X(\log X)^{-A}. \tag{100}$$

The masks may depend on X, M, Q and on the source row m, but not on the quotient j.

Proof. First assume $(m,h)=1$. Maximal Bombieri–Vinogradov and smooth partial summation give

$$\sum_{j\ge 1}\Lambda(mj+h)F(mj/X) = \frac{X}{\varphi(m)}I_0(F)+E_m, \tag{101}$$

$$\sum_{m\le M}|E_m| \ll_{A,h,F} X(\log X)^{-A}, \tag{102}$$

after increasing the requested exponent. On the other hand, lemmas 4.2 and 4.3 give

$$\sum_{j\ge 1}\Lambda_Q(mj+h)F(mj/X) = \frac{X}{\varphi(m)}I_0(F)+E^{(Q)}_m, \qquad |E^{(Q)}_m| \ll_F Q(\log(2Q))^2. \tag{103}$$

Thus the local main terms cancel row by row, not merely after the m sum.

If $(m,h)>1$, the common local mean is zero. A von Mangoldt value $\Lambda(mj+h)$ forces $mj+h$ to be a power of one of the finitely many primes dividing h. Summed over all $m\le M$, these exceptional rows contribute $O_{h,F,\varepsilon}(X^{\varepsilon})$: for each relevant prime power one counts only divisors of a fixed integer. The model rows still have mean zero by lemma 4.2, and

$$\sum_{j\ge 1}\Lambda_Q(mj+h)F(mj/X) = E^{(Q)}_m, \qquad |E^{(Q)}_m| \ll_F Q(\log(2Q))^2$$

by lemma 4.3.

Taking the supremum over the masks is now harmless. The preceding reduced and nonreduced row estimates give

$$\sup_{|\alpha_m|\le 1}\left|\sum_{m\le M}\alpha_m R_{m,h;Q}(X;F)\right| \ll_A X(\log X)^{-A}+MQ(\log(2Q))^2+X^{\varepsilon}. \tag{104}$$

The last two terms are absorbed by the power saving in (98).

For the logarithmic row, partial summation in maximal Bombieri–Vinogradov gives, when $(m,h)=1$,

$$\sum_{j\ge 1}(\log j)\Lambda(mj+h)F(mj/X) = \frac{1}{\varphi(m)}\int_0^\infty \log(t/m)F(t/X)\,dt+E^{\log}_m,$$

$$\sum_{m\le M}|E^{\log}_m| \ll_A X(\log X)^{-A}.$$

This main term equals the one in (83), because $\rho_h(m)/m = 1/\varphi(m)$. The prime part loses at most one power of $\log X$ in obtaining the displayed error. The model lattice error is (83); after summing over m it is $O(MQ(\log(2XQ))^3)$. Again the fixed power $X^{-\eta}$ absorbs every prescribed logarithmic factor. On nonreduced rows, the same prime-power argument as above costs only an additional factor $O(\log X)$, while the model has zero mean and obeys the same logarithmic lattice bound. This proves (100). $\square$

Corollary 6.2 (The power-separated calibration)**.** *Fix $0<\delta<1/2$ and take*

$$Q=\lfloor X^{1/2-\delta}\rfloor. \tag{105}$$

Then theorem 6.1 holds with $M\le Q$, since $MQ\le Q^2\le X^{1-2\delta}$.

This is a fixed-shift full dual-ball statement, but only in the row variable. Allowing a mask $\alpha_{m,j}$ that varies arbitrarily with the quotient would permit it to inspect $mj+h$ and select the target primes. Neither Bombieri–Vinogradov nor the theorem above permits that target-adaptive operation.

7 A Ramanujan-calibrated Vaughan reduction

We now isolate the precise packet not discharged by the preceding Type-I theorem. For $U, V \geq 1$, put

$$\beta_V(k) = \sum_{\substack{d \mid k \\ d \leq V}} \mu(d). \tag{106}$$

Vaughan's identity, with signs fixed for later use, is

$$\begin{aligned}\sum_t \Lambda(t)F(t) = \sum_{t\leq U} \Lambda(t)F(t) - \sum_{\substack{\ell \leq U \\ d\leq V}} \Lambda(\ell)\mu(d) \sum_{j\geq 1} F(\ell d j) \\ + \sum_{d\leq V}\mu(d)\sum_{j\geq 1}(\log j)F(dj) - \sum_{\substack{\ell > U \\ k>1}} \Lambda(\ell)\beta_V(k)F(\ell k).\end{aligned} \tag{107}$$

Moreover, $\beta_V(k)=0$ for $1<k\leq V$, so the last packet is supported on $\ell > U$, $k>V$. This is the classical identity [5, 11]; its use here is algebraic.

Fix $0<\delta<1/2$ and choose

$$Q = \left\lfloor X^{1/2-\delta} \right\rfloor, \qquad U = V = \left\lfloor Q^{1/2} \right\rfloor. \tag{108}$$

Then

$$UV \leq Q, \qquad (UV)Q \leq X^{1-2\delta}. \tag{109}$$

The first inequality ensures that every Type-I row sees all of its local Ramanujan factors. The second pays for the incomplete-period boundary errors.

Define the calibrated hard packet

$$\mathcal{H}^{\mathrm{HB}}_{h,Q;U,V}(X) = \sum_{\substack{\ell > U \\ k>V}} \Lambda(\ell)\beta_V(k) r_Q(\ell k + h) W(\ell k/X). \tag{110}$$

For comparison with the sign convention in the introduction, set

$$\mathcal{B}_{h,\delta}(X) := -\mathcal{H}^{\mathrm{HB}}_{h,Q;U,V}(X). \tag{111}$$

Theorem 7.1 (Exact Type-II residual reduction)**.** *Let $h\neq 0$ be fixed and $W \in C_c^\infty((0,\infty))$. With the parameters in (108), for every $A>0$,*

$$\begin{aligned}&\sum_t \Lambda(t)\Lambda(t+h)W(t/X) \\ &\quad = \mathfrak{S}(h)XI_0(W) - \mathcal{H}^{\mathrm{HB}}_{h,Q;U,V}(X) + O_{A,h,W,\delta}(X(\log X)^{-A}).\end{aligned} \tag{112}$$

Proof. Split the target weight as $\Lambda(t+h) = \Lambda_Q(t+h) + r_Q(t+h)$. The first part is evaluated by theorem 5.2 and lemma 5.1:

$$\sum_t \Lambda(t)\Lambda_Q(t+h)W(t/X) = \mathfrak{S}(h)XI_0(W) + O_A(X(\log X)^{-A}). \tag{113}$$

Apply (107) to

$$F(t) = r_Q(t+h)W(t/X). \tag{114}$$

Because W is supported in a compact subinterval of $(0,\infty)$ and $U = o(X)$, the first packet $t \le U$ is identically zero for all sufficiently large X.

For the second packet, group by $m = \ell d$. Its row coefficient is

$$A_{U,V}(m) = \sum_{\substack{\ell d = m\\ \ell \le U, d \le V}} \Lambda(\ell)\mu(d), \qquad |A_{U,V}(m)| \le \sum_{\ell | m} \Lambda(\ell) = \log m. \tag{115}$$

Every such m satisfies $m \le UV \le Q$. Divide the coefficient by $\log X$, apply (99), and request one additional logarithmic power. The entire second packet is therefore $O_A(X(\log X)^{-A})$.

The third packet is exactly a bounded mask of the logarithmic rows (97), with $d \le V \le Q$. Equation (100) gives the same error. The non-reduced rows in both packets are already included in theorem 6.1; their von Mangoldt support consists only of powers of the finitely many primes dividing h.

Finally, (106) vanishes for $1 < k \le V$, so the last term in Vaughan's identity is $-\mathcal{H}^{\mathrm{HB}}_{h,Q;U,V}(X)$. Combining it with (113) proves (112). □

The reduction can also be compared directly with the global residual correlation.

Corollary 7.2 (Hard packet versus residual autocorrelation)**.** *Under the hypotheses of theorem 7.1,*

$$\begin{aligned}\mathcal{H}^{\mathrm{HB}}_{h,Q;U,V}(X) = &-\sum_n r_Q(n) r_Q(n+h) W(n/X)\\ &+ O_{A,h,W,\delta}(X(\log X)^{-A}).\end{aligned} \tag{116}$$

Proof. Compare (112) with (95). □

Corollary 7.3 (The fixed-shift Type-II gate)**.** *Assume that h is an admissible even shift and that $I_0(W) \ne 0$. Then*

$$\sum_t \Lambda(t)\Lambda(t+h)W(t/X) \sim \mathfrak{S}(h) X I_0(W) \quad \Longleftrightarrow \quad \mathcal{H}^{\mathrm{HB}}_{h,Q;U,V}(X) = o(X). \tag{117}$$

Proof. This is an immediate consequence of (112), since its remainder is $o(X)$. □

On the support of the hard packet, $\ell k \asymp_W X$ and both factors exceed $X^{1/4-\delta/2}$; hence each is at most $O_W(X^{3/4+\delta/2})$. This is a genuine broad Type-II region. The theorem identifies it but does not estimate it. In particular, (117) is not evidence that the right-hand condition holds for $h = 2$. The use of r_Q, rather than the raw von Mangoldt target, is also essential: asking the uncalibrated hard packet to be $o(X)$ would generally subtract the wrong local main term.

8 Boundary certificate: what the reduction does and does not prove

This section records the logical endpoint of the argument. It separates three statements which have different observables, different natural scales, and different quantifiers: the row-centered primitive-shell residual, one prescribed prime-pair shift, and a finite bundle of candidate shifts. None may be substituted for another without an additional theorem.

8.1 The shell and the endpoint live on different scales

For fixed $h \neq 0$, put

$$\mathcal{P}_h(X) := \sum_n \Lambda(n)\Lambda(n+h)W(n/X).$$

The fixed-shift upper-bound sieve gives

$$\mathcal{P}_h(X) \ll_{h,W} X, \tag{118}$$

with the prime-power terms absorbed in the same bound; see, for example, Iwaniec and Kowalski [7]. The Hardy–Littlewood prediction is the much sharper asymptotic

$$\mathcal{P}_h(X) = \mathfrak{S}(h) X I_0(W) + o(X), \tag{119}$$

for admissible even h [4]. Thus the endpoint problem has natural scale X.

By contrast, the complete row-centered primitive-shell residual proved above has the form

$$\mathcal{R}_h(X) = \left(\frac{\varphi(|h|)}{|h|} - \beta_h\right) X I_0(W) \log X + O_{h,W}(X). \tag{120}$$

When $\delta_h I_0(W) \neq 0$, its scale is $X \log X$. This extra logarithm is factor-coordinate multiplicity, not an extra logarithm in the endpoint correlation. Moreover, $\mathcal{R}_h$ is a signed, centered sum over primitive factorizations, whereas $\mathcal{P}_h$ is a one-dimensional endpoint observable, nonnegative only when $W \geq 0$. No positivity-preserving comparison between the two quantities has been proved. In particular, dividing (120) by $\log X$ does not yield (119); even the positive coefficient $1/6$ when $h = 2$ is not a twin-prime lower bound. This is the same label-preservation boundary isolated in TPC-10 and TPC-14 [13, 14].

8.2 A finite selector theorem

There is nevertheless a rigorous existence mechanism when the shift bundle is fixed and finite. It is useful to state it independently of the primes.

Theorem 8.1 (Finite selector)**.** *Let $K \geq 2$ be fixed and let $x(n) = (x_1(n), \ldots, x_K(n)) \in \{0,1\}^K$. Define*

$$E(N) := \sum_{n \leq N} \mathbf{1}_{\{\sum_i x_i(n) \geq 2\}}, \qquad C_{ij}(N) := \sum_{n \leq N} x_i(n)x_j(n) \quad (i < j).$$

Then, for every N,

$$E(N) \leq \sum_{i<j} C_{ij}(N), \qquad \max_{i<j} C_{ij}(N) \geq \frac{E(N)}{\binom{K}{2}}. \tag{121}$$

If $E(N) \to \infty$, there are a fixed pair (i_*, j_*) and a sequence $N_r \to \infty$ for which

$$C_{i_*j_*}(N_r) \geq \frac{E(N_r)}{\binom{K}{2}}.$$

In particular, $C_{i_*j_*}(N) \to \infty$.

Proof. For each n,

$$\mathbf{1}_{\{\sum_i x_i(n) \geq 2\}} \leq \sum_{i<j} x_i(n)x_j(n).$$

Summing gives the first inequality in (121), and averaging over the $\binom{K}{2}$ pairs gives the second. Choose a maximizing pair for each N. One pair occurs for infinitely many N because the set of pairs is finite. Passing to that subsequence proves the quantitative assertion. Since $E(N_r) \to \infty$ and each $C_{ij}(N)$ is nondecreasing, the last conclusion follows. $\square$

In symbolic-dynamical language, the set of states with at least two active coordinates is the finite union

$$\left\{x : \sum_i x_i \geq 2\right\} = \bigcup_{i<j} \{x : x_i = x_j = 1\}.$$

Infinite recurrence to this union therefore forces infinite recurrence to at least one fixed two-coordinate cylinder. Finiteness is essential: the argument does not invert a normalized average over a growing family of shifts.

Corollary 8.2 (Standard bounded-gap consequence at an unidentified shift)**.** *There exists a fixed even integer h_* with*

$$2 \leq h_* \leq 246$$

such that p and $p + h_*$ are simultaneously prime for infinitely many primes p.

Proof. This is the Polymath8b bounded-gap theorem followed by finite pigeonhole; it is not a new bounded-gap theorem and is not used in the analytic reduction above. Specifically, Theorems 16(i) and 17 of D. H. J. Polymath [2] give DHL[50; 2] and $H(50) = 246$. Fix an admissible 50-tuple

$$\mathcal{H} = \{h_1 < \cdots < h_{50}\} \subset [0, 246]$$

of diameter 246, and apply theorem 8.1 to $x_i(n) = \mathbf{1}_{\mathbb{P}}(n + h_i)$. The DHL[50; 2] assertion says that $E(N) \to \infty$, so one fixed pair (i_*, j_*) is coactive infinitely often. Take $h_* = h_{j_*} - h_{i_*}$. Admissibility modulo 2 forces all h_i to have the same parity; hence h_* is even, and its diameter gives $2 \leq h_* \leq 246$. □

The corollary is an exact prime–prime recurrence statement, but its quantifier is "there exists h_*". It neither identifies h_* nor permits the substitution $h_* = 2$.

8.3 Why the aggregate cannot identify the coordinate

The loss of the shift label is already exact in the abstract finite-channel model.

Proposition 8.3 (Aggregate non-identifiability)**.** *Let $K \geq 3$. Knowledge of the entire aggregate path $\{E(N) : N \geq 1\}$, even together with $\{\sum_{i<j} C_{ij}(N) : N \geq 1\}$, does not determine which pair is coactive infinitely often on the class of binary K-channel sequences.*

Proof. Fix an infinite set $S \subset \mathbb{N}$ and two distinct coordinate pairs $e \neq f$. In the first sequence set precisely the two coordinates in e equal to 1 when $n \in S$, and set every coordinate to 0 otherwise. Construct the second sequence in the same way with f in place of e. For both sequences and every N,

$$E(N) = \sum_{i<j} C_{ij}(N) = |S \cap [1, N]|,$$

but their unique recurrent coactivation pairs are different. □

This proposition is an information statement about the aggregate observable, not a claim of logical independence for the actual primes. Recovering a prescribed shift requires label-sensitive input that is absent from the aggregate.

8.4 The unresolved packet

The Ramanujan calibration and Vaughan decomposition above reduce the endpoint error to the displayed balanced Type-II packet, up to terms already shown to be $o(X)$. Recall that $\mathcal{B}_{h,\delta} = -\mathcal{H}^{\mathrm{HB}}_{h,Q;U,V}$. The present paper does not prove the required estimate

$$\mathcal{H}^{\mathrm{HB}}_{h,Q;U,V}(X) = o(X) \tag{122}$$

for that packet. Consequently it does not prove (119), a positive lower bound for $\mathcal{P}_2(X)$, or the twin-prime conjecture. The decomposition is a precise information gate: Type-I distribution and the finite Ramanujan spectrum have been discharged, while genuinely balanced, label-sensitive bilinear control remains. Known parity-barrier analyses explain why such additional information cannot be replaced by formal sieve rearrangement alone [3, 10].

For clarity, the final ledger is therefore:

Object	Natural scale	Status here
Complete row-centered residual	$X \log X$	Explicit asymptotic; still summed over all primitive orientations.
Fixed-shift prime endpoint	X	Upper-bound scale only; no Hardy–Littlewood asymptotic.
Finite admissible shift bundle	recurrence	Some fixed even $h_* \le 246$ is selected; its identity is not recovered.
Balanced hard packet	X target	Isolated exactly; the necessary $o(X)$ estimate is not proved.

9 Conclusion and the next gate

The present fixed-shift reduction has three rigorously separated layers.

First, the primitive shell admits a row-local prime density. Below the quarter scale, maximal Bombieri–Vinogradov controls every bounded source-factor row mask after this centering. The complete row-centered residual still has coefficient δ_h, equal to $1/6$ at $h = 2$. When $\delta_h I_0(W) \neq 0$, its logarithmic main term is not an artifact of using the wrong mean; it is forced into the post-quarter aggregate.

Second, the finite Heath–Brown model realizes the local factors as an actual translation spectrum. Its rational eigenfrequencies are finite, its autocorrelation is the truncated singular series, and it matches the two prime–model cross terms required here through Bombieri–Vinogradov. Subtracting it removes the Hardy–Littlewood local main term without assuming the Hardy–Littlewood conjecture.

Third, the Ramanujan residual is calibrated on the full quotient-independent Type-I row ball used here, up to $Q = X^{1/2-\delta}$. Vaughan's identity therefore leaves one explicit hard packet with both factor variables beyond the Type-I cutoff. The logical endpoint is exact:

$$\text{weighted Hardy–Littlewood at admissible even } h, I_0(W) \neq 0 \quad \Longleftrightarrow \quad \mathcal{B}_{h,\delta}(X) = o(X).$$

This equivalence is the result of the reduction, not an estimate asserted by it.

The next analytic target is correspondingly narrow. One may seek a power- or logarithmic-saving estimate for dyadic pieces of $\mathcal{B}_{h,\delta}$ after retaining the Ramanujan subtraction, perhaps first under an additional average over one factor, a sparse set of shifts, or a restricted family of bilinear masks. Any such theorem must be checked at the natural X scale and against the correct singular-series center. Failure at that stage would itself locate which bilinear corridor or mask class supports the obstruction.

No argument here proves a fixed-shift prime-pair asymptotic, a twin-prime lower bound, a level of distribution beyond one half, or a breach of the sieve parity barrier. The advance is a sharper interface: the local factor is explicit, the required Type-I row ball is closed at a prescribed shift, and the remaining fixed-shift task is a single named bilinear residual rather than an unspecified appeal to "more cancellation."

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